

XMBA200S: Data and Decisions

Knowledge Check 1

UC Berkeley – Haas School of Business

Production Capacity for a Meal-Kit Company

A meal-kit company is planning production for a new product. Weekly demand varies from week to week and is approximately normally distributed:

$$D \sim N(10,000, 2,000^2),$$

where D is the number of meal kits demanded in a week.

Each meal kit requires 0.40 hours of packaging-line time. Therefore, the total number of packaging hours required in a week, H , is

$$H = 0.40D.$$

Management must decide how much weekly production capacity to install.

[Total points: 20 pts]

Question 1: Packaging Requirements [2 pts]

Find the mean and standard deviation of weekly packaging hours H .

- Mean: _____ hours
- Standard deviation: _____ hours

Solution. Since

$$H = 0.40D$$

and

$$D \sim N(10,000, 2,000^2),$$

we use the rules for a linear transformation.

The mean is

$$E[H] = 0.40E[D] = 0.40(10,000) = \boxed{4,000}.$$

The standard deviation is

$$SD(H) = 0.40SD(D) = 0.40(2,000) = \boxed{800}.$$

Thus,

$$\boxed{H \sim N(4,000, 800^2)}.$$

Question 2: Capacity at Average Demand [2 pts]

Management initially proposes installing 4,000 hours of weekly capacity because this equals the average weekly requirement. What is the probability that required packaging hours exceed available capacity in a randomly selected week?

- A. 15.9%
- B. 25.0%
- C. 50.0%
- D. 84.1%

Solution. Since

$$H \sim N(4,000, 800^2),$$

we want

$$P(H > 4,000).$$

Standardizing,

$$P(H > 4,000) = P\left(Z > \frac{4,000 - 4,000}{800}\right) = P(Z > 0).$$

Therefore,

$$P(H > 4,000) = 0.50.$$

The correct answer is

$$\boxed{\text{C. 50.0\%}}.$$

Installing capacity equal to average demand means that demand will exceed capacity in approximately half of all weeks.

Question 3: Evaluating a Larger Capacity [3 pts]

The operations team instead proposes installing 4,800 hours of weekly packaging capacity. What is the probability that required packaging hours exceed this capacity in a randomly selected week? Enter your answer as a percentage, rounded to one decimal place:

_____ %.

Solution. We want

$$P(H > 4,800).$$

Using

$$H \sim N(4,000, 800^2),$$

we standardize:

$$P(H > 4,800) = P\left(Z > \frac{4,800 - 4,000}{800}\right) = P(Z > 1).$$

From the standard normal table,

$$P(Z < 1) = 0.8413.$$

Therefore,

$$P(Z > 1) = 1 - 0.8413 = 0.1587.$$

Thus,

$$P(H > 4,800) \approx 15.9\%.$$

Question 4: Planning for a Multi-Week Promotion [2 pts]

The company is planning a 4-week promotional campaign. During the campaign, management wants to avoid any production delays, which means that the 4,800 hours of capacity must be sufficient in each of the four weeks. From the previous question, the probability that 4,800 hours are sufficient in any given week is approximately 84.1%.

Assume weekly demands are independent. Which expression gives the probability that the company has sufficient capacity throughout the entire 4-week promotion?

- A. 0.841
- B. $(0.841)^4$
- C. $1 - (0.841)^4$
- D. $1 - (0.159)^4$

Solution.

The correct answer is

$$\text{B. } (0.841)^4.$$

The probability 0.841 refers to having sufficient capacity in one week. For the company to have sufficient capacity throughout the promotion, capacity must be sufficient in week 1 **and** week 2 **and** week 3 **and** week 4. Since weekly demands are independent,

$$P(\text{sufficient capacity in all 4 weeks}) = (0.841)^4 \approx 0.500.$$

Thus, the probability of having sufficient capacity throughout the entire 4-week promotion is approximately

$$50.0\%.$$

Question 5: Required Weekly Demand Capacity [8 pts]

Management wants enough capacity so that demand exceeds available capacity in only 5% of weeks.

Question 5.1 [2 pt]

Let d^* denote the weekly demand level the company plans to accommodate. Which probability statement represents management's objective?

- A. $P(D \leq d^*) = 0.05$
- B. $P(D > d^*) = 0.05$
- C. $P(D > d^*) = 0.95$
- D. $P(D = d^*) = 0.05$

Solution.

The correct answer is

$B. P(D > d^*) = 0.05$.

Management is willing to have demand exceed available capacity in only 5% of weeks.
Equivalently,

$$P(D \leq d^*) = 0.95.$$

Question 5.2 [2 pt]

Using your answer from Question 5.1 and the standard normal table, what is the corresponding z -score?

$$z = \underline{\hspace{2cm}}$$

Solution.

From Question 5.1, management wants to accommodate 95% of the weekly demand distribution. The corresponding value from the standard normal table is

$z \approx 1.645$.

Question 5.3 [2 pts]

Using your answers from Questions 5.1 and 5.2, what weekly demand should the company plan to accommodate? Enter your answer in number of meal kits:

Solution.

From Question 5.1, management wants demand to exceed available capacity in only 5% of weeks:

$$P(D > d^*) = 0.05.$$

Equivalently,

$$P(D \leq d^*) = 0.95.$$

From Question 5.2, the z -score corresponding to this probability is

$$z = 1.645.$$

We therefore standardize d^* :

$$P\left(Z \leq \frac{d^* - 10,000}{2,000}\right) = 0.95.$$

Since the z -score associated with a cumulative probability of 0.95 is 1.645,

$$\frac{d^* - 10,000}{2,000} = 1.645.$$

Solving for d^* ,

$$d^* = 10,000 + 1.645(2,000) = \boxed{13,290}.$$

Thus, the company should plan to accommodate approximately 13,290 meal kits per week.

Question 5.4 [2 pts]

Each meal kit requires 0.40 packaging hours. Using your answer from Question 5.3, how many weekly packaging hours should the company install?

_____ hours.

Solution. The company needs capacity for approximately 13,290 meal kits. Therefore,

$$H^* = 0.40(13,290) = 5,316.$$

Thus, the required weekly packaging capacity is approximately

$$\boxed{5,316 \text{ hours}}.$$

Question 6: Management Decision [3 pts]

Management is considering two alternatives:

- **Option A:** 4,800 hours of weekly capacity;
- **Option B:** 5,316 hours of weekly capacity.

Which of the following statements are correct? **Select all that apply.**

- A. Option A requires less installed capacity than Option B.
- B. Option A may still be preferable if the cost of additional capacity is high relative to the cost of occasionally not meeting demand.
- C. Moving from Option A to Option B eliminates most of the uncertainty in weekly demand
- D. If failing to meet demand is very costly, Option B becomes relatively more attractive.

Solution. The correct answers are

A, B, and D.

A is correct. Option A requires 4,800 hours of weekly capacity, compared with 5,316 hours under Option B.

B is correct. Option B reduces the risk of insufficient capacity, but achieving this lower risk requires additional capacity, which may be costly. If the cost of adding capacity is high relative to the cost of occasionally not meeting demand, management may rationally prefer Option A despite its higher shortage risk.

C is incorrect. Increasing capacity does not reduce the uncertainty in weekly demand. The demand distribution remains the same under both options. Option B only reduces the probability that demand exceeds available capacity.

D is correct. If failing to meet demand is very costly, reducing the probability of insufficient capacity becomes more valuable. Since Option B lowers this probability from approximately 15.9% to 5%, it becomes relatively more attractive when shortage costs are high.